

# Leggett-Garg Inequality-based Quantum Computers Quality Benchmark

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# Motivation

- Need for independent benchmarking tools
- With accordance to European Quantum Strategy
- The use of Star-topology devices in the advent of Constellation-type devices
- ...

# Executive summary

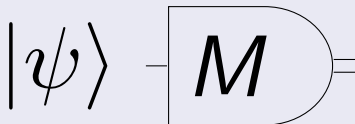
Executive summary ( judging panel made up of Department Heads, Management, and VPs of Technology)

- What we did
- Additional technical results (Star Device Qubit Selector, FakeSirius)

# Projective vs weak measurement

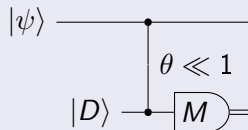
## Projective (strong) measurement

- Direct
- Accurate (in theory)
- Disturbing (decoherence)



## Weak measurement

- Indirect (weak coupling)
- Inaccurate (deliberately)
- Small decoherence



## No free lunch

- Information gain  $> \theta$
- Decoherence  $> \theta^2$

## The catch: Measurement strength $\theta$ calibration

$$a_{\text{raw}} = \theta a_{\text{physical}}$$

# Application: Leggett-Garg inequality violation

## Leggett-Garg (LG) Inequality

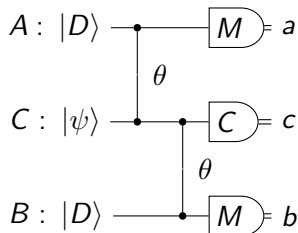
$$\langle a \rangle + \langle b \rangle - \langle ab \rangle \leq 1, |a| \leq 1, |b| \leq 1$$

## LG-based benchmark inequality

$$Q = \frac{\langle a|c \rangle \langle b|c \rangle}{\langle ab|c \rangle} \leq 1,$$

where  $\langle ab|c \rangle = \langle abc \rangle / p(c = 1)$ .

- measurement strength-agnostic
- postselection  $c = 0, 1$ ,
- $A \equiv B$  (identical, independent)



## Detectors identity and independence ( $A \equiv B$ )

Partially verifiable by  $D = 0$ , where  $D := \langle a \rangle \langle bc \rangle - \langle b \rangle \langle ac \rangle$ .

# A test with strength dependence on $\lambda = \sin \theta$

## Setup

$$A = B = Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

$$C = |\psi_-\rangle\langle\psi_-|$$

$$|\psi\rangle = |\psi_+\rangle$$

$$|\psi_{\pm}\rangle = \cos \frac{\psi}{2} |1\rangle \pm \sin \frac{\psi}{2} |0\rangle$$

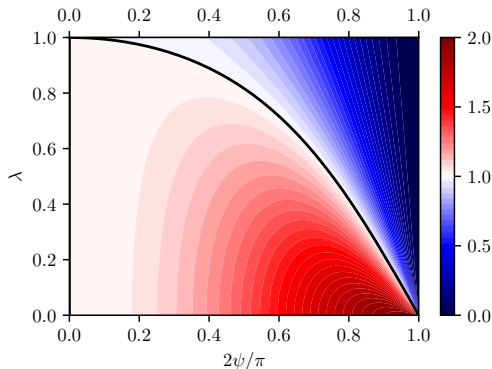


Figure: Dependence of  $Q$  on  $\psi$ .

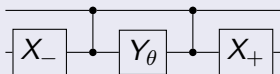
# Weak coupling implementation

## Outline

- Two qubit gate  $U \sim I$
- Basis:  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

## Weak coupling with IQM-native gates

We use two  $CZ$  gates sandwiching  $Y_\theta$  gate.



## Notation

$$X = |0\rangle\langle 1| + |1\rangle\langle 0|$$

$$Y = i|1\rangle\langle 0| - i|0\rangle\langle 1|$$

$$V_\theta = \exp(-i\theta V)$$

$$V_\pm = V_{\pm\pi/2}$$

## Alternative: Fractional gates

Weak coupling implementation is native on IBM / IonQ devices with the fractional RZZ gate.

$$RZZ(\theta) = \text{diag}(1, e^{i\theta}, e^{i\theta}, 1)$$

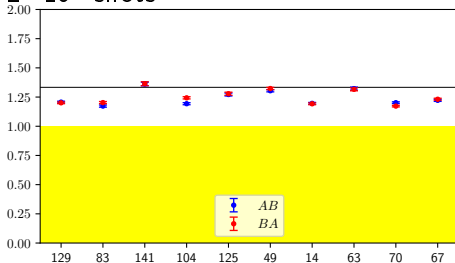
Detectors independence results



# Results on IBM/IonQ/IQM $\theta = 0.1$

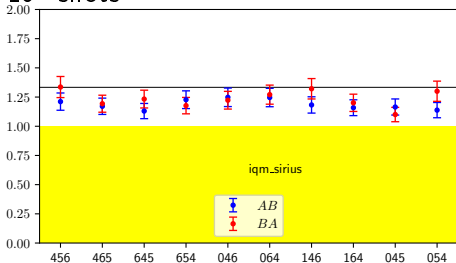
IBM kingston ( $C$  qubit specified)

$2 \cdot 10^7$  shots



IQM Sirius ( $CAB$  qubits specified)

$10^6$  shots



IonQ Forte 1:  $1.272 \pm 0.035$  ( $A \rightarrow B$ ),  $1.289 \pm 0.035$  ( $B \rightarrow A$ )

Proposed benchmark results?

IQM Pulla failed results

# Impact and roadmap

## Challenges

- more access needed to lower error  $\sigma \sim 1/\theta^2 \sqrt{N}$ ,
- pulse control  $U \rightarrow U^\theta$ , e.g.  $CZ \rightarrow \text{diag}(1, 1, 1, e^{\pm i\theta})$ , continuous families of entangling gates
- avoid postselection (full noninvasiveness)
- Star topology  $\rightarrow$  scale up the test, more detectors
- TODO(TR): CZ protocol works better!

## Benefits

- benchmarking partially entangling gates and pulse control
- proposed benchmarking score/map (for the control range)

$$Q - 1, (Q - 1)/\sigma$$

- minimal disturbance:  $\theta = 1/10$  disturbance  $1/400$ .
- robustness to noise – no error correction needed

Scalability

# Thank you for your attention!

[https://github.com/Tomev/IQM\\_Research\\_Award\\_2026](https://github.com/Tomev/IQM_Research_Award_2026)

## **Last but not least! Thanks to:**

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